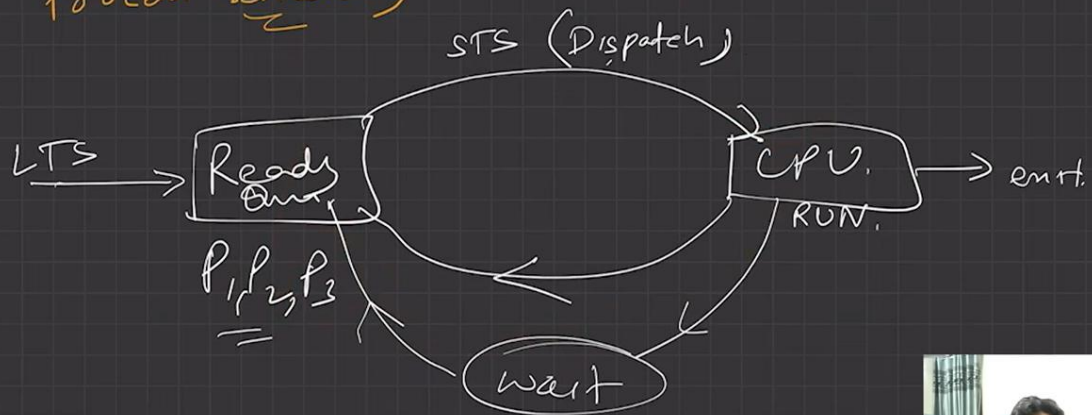
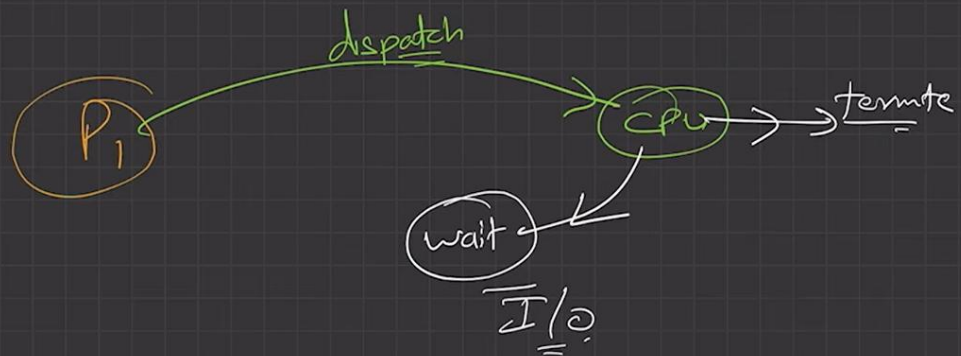


# Lec-12

## Process scheduling



## ① Non-preemptive scheduling :-



## ② Preemptive sched

time-quantum

① Starvation  $\rightarrow$  Non-preemptive  $\uparrow$   
Preemptive  $\downarrow$

② CPU utilization  $\rightarrow$  Preemptive  $\uparrow$

③ Overhead  $\rightarrow$  Preemptive  $\uparrow$   $\checkmark$

\* Goals of CPU scheduling :-

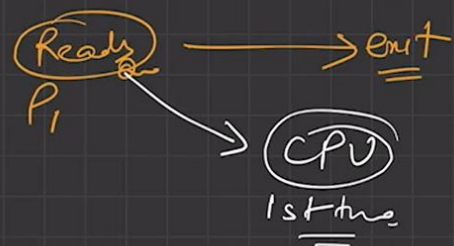
① Max. CPU utilization :-

② Min. TAT

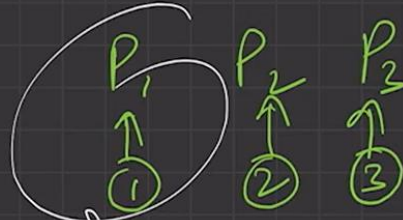
③ Min. WT

④ Min. Response time

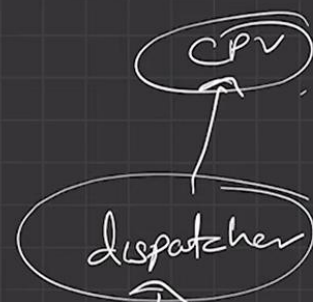
⑤ Max. Throughput  
 $\downarrow$



① FCFS



Selected first



## Convoy effect :-

|   | P <sub>no</sub> | AT | BT | CT | (CT-AT)<br>TAT | (TAT-BT)<br>WT |
|---|-----------------|----|----|----|----------------|----------------|
| I | 1               | 0  | 20 | 20 | 20             | 0              |
|   | 2               | 1  | 2  | 22 | 21             | 19             |
|   | 3               | 2  | 2  | 24 | 22             | 20             |
|   |                 |    |    |    | Avg WT 13      |                |

Gantt Chart  


|   | P <sub>no</sub> | AT | BT | CT | TAT         | WT |
|---|-----------------|----|----|----|-------------|----|
| I | P <sub>2</sub>  | 0  | 2  | 2  | 2           | 0  |
|   | P <sub>3</sub>  | 1  | 2  | 4  | 3           | 1  |
|   | P <sub>1</sub>  | 2  | 20 | 24 | 22          | 2  |
|   |                 |    |    |    | Avg. 1 unit |    |

